Quack Stack TM

Today's Agenda...

01 Introduction

02 Phylosophy Of Juan Pato

03 Meaningful Atoms

04 Friends Of Order

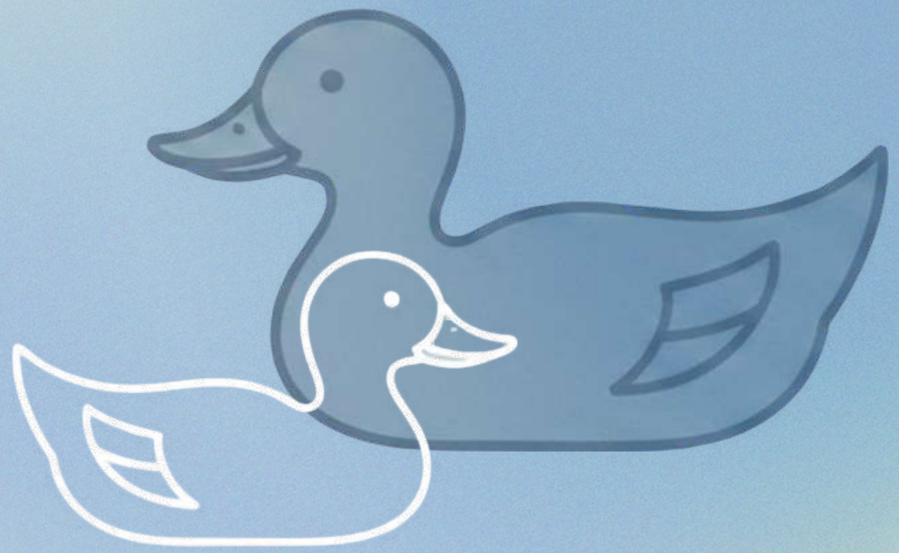
05 Not A Transator, A Gatekeeper

06 The Road to innovation

Presented by :

Quack Stack TM

UNAM ENGINEERING FACULTY



Juan pato v0.1

The architecture of meaning

AMBIGUITY

GRAMMAR (G)

E → E + E | E * E | (E) | id
S → if E thenS else S | if E thenS
S → while E do S | { L }
L → L ; S | S

```
if a < b then
   if c < d then
      x = y + z * w
   else
      x = y * z + w
```

Which 'if' does else match?

PARSER TABLE (LR(1))

State	id	+	(	)	$	Action
5		S5				...
6		r2				r2 / s7
						r4 / s8
						...

Syntax Error
Unexpected token 'else' in input
OK

Syntax Error
Unexpected token '*'
Expected ')'
OK

```
int func(int a) {
  if (a < 10)
     return a * a + 1;
  else
     return a * (a + 1);
```

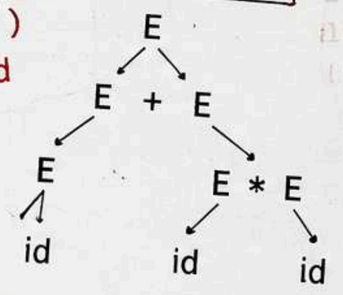
Syntax Error
Reduce/Reduce conflict in state 12
OK

```
x = a + b * c + d
```

(a + b) * (c + d) ?
a + (b * (c + d)) ?
(a + (b * c)) + d ?

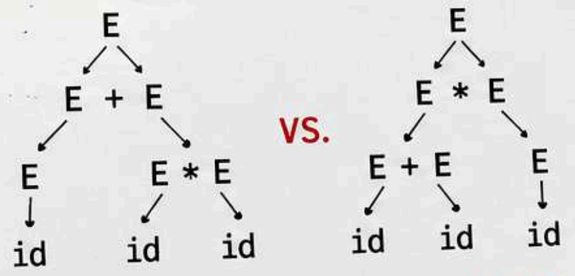
E → E + E
E → E * F
E → (E)
E → id

Ambiguous Grammar



Multiple Parse Trees!

ALL POSSIBLE PARSE TREES



Same input, different trees, different meanings!

```
while i < n do
  if a[i] == 0 then
     i = i + 1
  else
     break
```

Dangling else problem!

Syntax Error
Ambiguous grammar: multiple parses possible
OK

AMBIGUITY
Leads to:
- Shift/Reduce Conflicts
- Reduce/Reduce Conflicts
- Uncertain Parse Trees
- Wrong Meaning!

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Phylosophy of Juan Pato

“Syntactic precision, visionary execution”

- Leonardo Leyva, CEO

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A language designed for clarity and safety

```
ABSTRACT SYNTAX TREE:
←[93m
              program

      _________|_____

   write              int

     |            _____|___

"This is an     a         10

  example"
```

```
-----------------------------------
Starting semantic analysis...
-----------------------------------
←[0m
←[92mSDT Verified!←[0m
```

```
Total tokens identified: 13

Lexical analysis completed.
```

```
Starting syntactic analyzer...
-----------------------------------
←[0m
←[92m
Parsing Success!←[0m
```

PIPELINE

LEXICAL

Breaking code into tokens

Converts a stream of characters into meaningful tokens.

if, (, id, <, num,), {, ...

→

SYNTACTIC

Checking structure

Verifies that the token sequence follows the grammar rules.

Parse Tree / AST

→

SEMANTIC

Understanding meaning

Analyzes meaning, types, and context to ensure the program makes sense.

Symbol Table / Type Info / Checks

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Meaningful Atoms

How does the compiler transform a sequence of characters into tokens?

INTEGER = (– | ε) D +

```
Tokens Identified:

TOKEN TYPE          VALUE
----------------------------------------
START               Start
L_BRACKET           {
WRITING_OP          printf
L_PARENTHESYS       (
STRING              "This is an example"
R_PARENTHESYS       )
SEMI_COLON          ;
TYPE                int
ID                  a
ASIG_OP             =
INTEGER             10
SEMI_COLON          ;
R_BRACKET           }
----------------------------------------
```

Friends Of Order

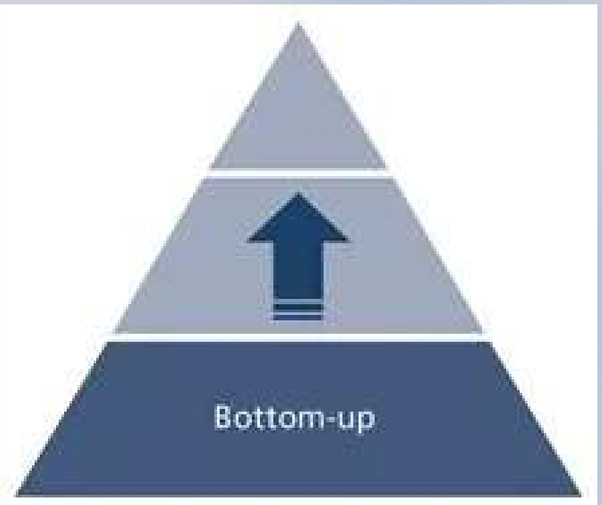
PARSER: WHAT IS IT?

Converts raw tokens into an organized hierarchy, a structure called Abstract Syntax Tree (AST).

```
ABSTRACT SYNTAX TREE:
←[93m
            program

     _________|____

   write            int

     |          ____|__

"This is an     a        10

  example"
```

BOTTOM UP

Provides a greater flexibility by building the logic, from specific details (leaves), to the general structure (root).




```
precedence=[
    ('left', ['BOOL_OP']),
    ('left', ['COMP_OP']),
    ('left', ['ARIT_OP']),
])
```

PRECEDENCE ORDER

Resolves ambiguity and ensures the parser builds the logical hierarchy required for correct evaluations.

Our Grammar is...

A Context Free Grammar

```
@self.pg.production('program : START L_BRACKET instructions R_BRACKET')
@self.pg.production('instructions : instruction instructions')
@self.pg.production('instructions : instruction')
```

And Recursive

```
@self.pg.production('instructions : instruction instructions')
@self.pg.production('expression : expression ARIT_OP expression')
@self.pg.production('expression : expression COMP_OP expression')
@self.pg.production('expression : expression BOOL_OP expression')
```

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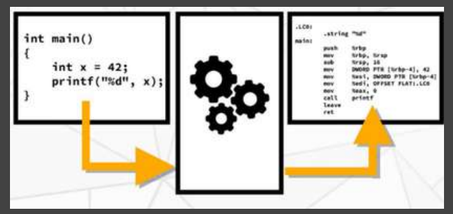
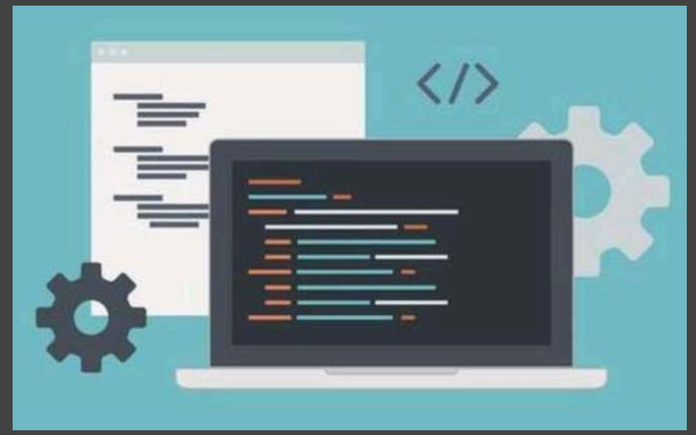


Not a Translator, a Gatekeeper



PERFECT LOGIC GOES INTO PRODUCTION

TWO TOOLS

- THE TREE

- THE SYMBOL TABLE

ERRORS

- SYNTACTIC
- SEMANTIC

The Road to innovation

Version 1 — **Adoption** — **Version 2** — **Expansion**

Version 1 (Academic Edition)

- Free access for universities
- Focused on learning and experimentation
- Easy to use and understand

Adoption Phase

- Students and institutions start using it
- Familiarity and trust are built
- Becomes part of the learning process

Version 2 (Professional Edition)

- Paid solution
- Modular design
- Adaptable to real-world needs

Expansion

- Industry use
- Continuous updates
- Professional support and maintenance

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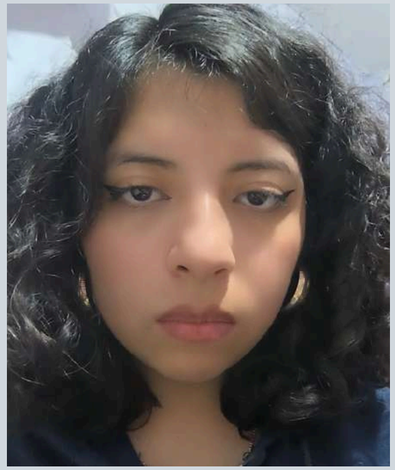
Meet Our Team

KARINA ROSETE
LEAD ENGINEER

NILSEN FLORES
ENGINEER



SARA LOPEZ
ENGINEER

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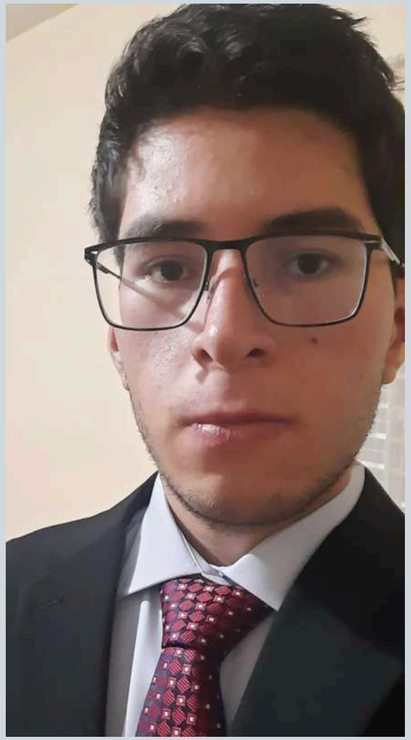
Meet Our Team



LEONARDO LEYVA

CEO

VICTOR

PROJECT MANAGER

Presented by :
Quack Stack Team

Thank You

for your time and attention